

AI Enterprise Knowledge Manager

Multi-Agent RAG System for Enterprise Knowledge Retrieval

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GitHub: <https://github.com/Ayushraina69/AI-Enterprise-Knowledge-Manager>

NOVATECH SOLUTIONS

FINAL YEAR AI PROJECT

The Problem: Enterprise Knowledge is Fragmented

Information Silos

Company knowledge is scattered across policies, meeting notes, SOPs, and project documents — with no unified access point.

Inefficient Search

Traditional keyword search fails to understand natural-language intent, wasting employee time on every query.

No Intelligent Interface

Employees lack a single intelligent system to ask questions and receive grounded, accurate answers from internal sources.



Objective & Proposed Solution

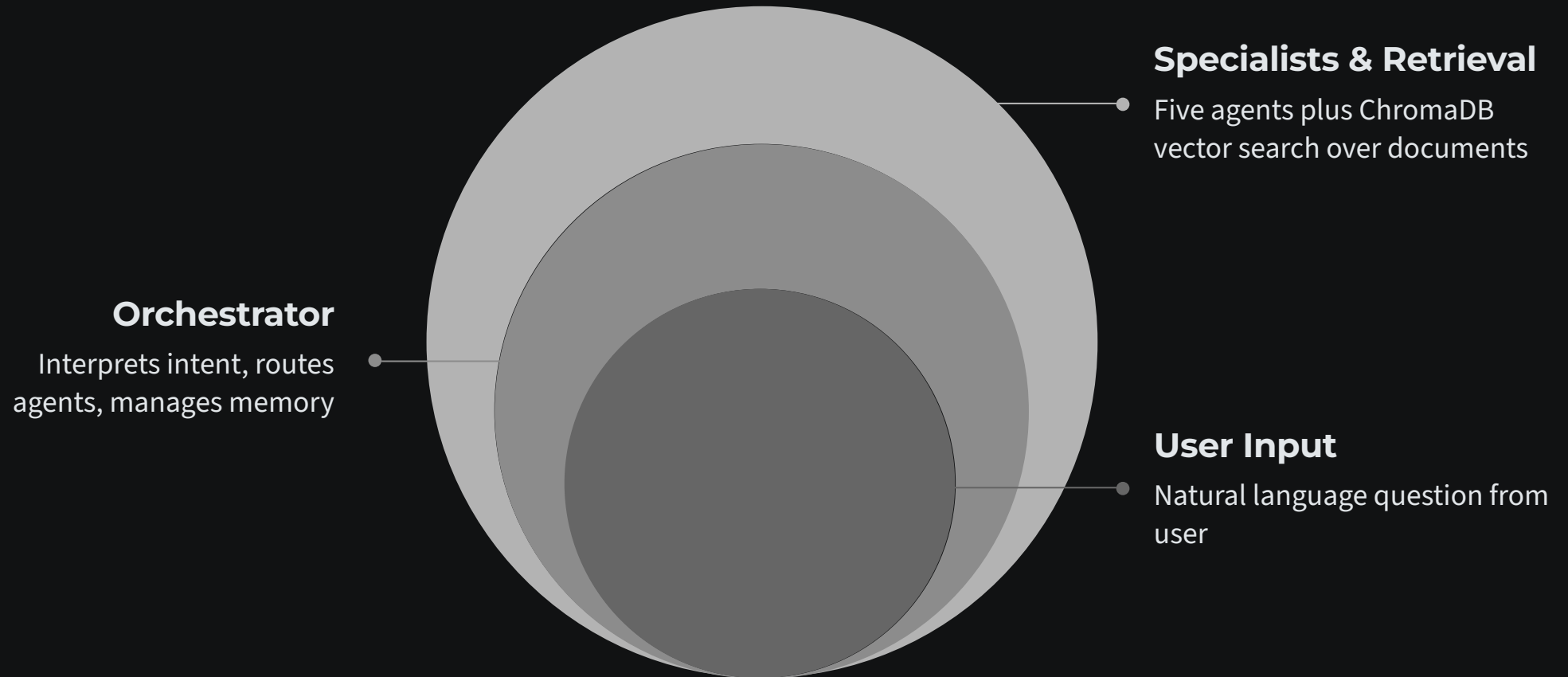


AI Enterprise Knowledge Manager

An AI-powered system that lets employees ask natural-language questions and receive accurate, grounded answers from internal company knowledge — powered by a multi-agent architecture and RAG.

- Natural-language enterprise knowledge retrieval
- Intelligent question routing to specialist agents
- Grounded answers sourced from internal documents
- Conversation memory for contextual continuity
- Reduces manual information searching

System Architecture

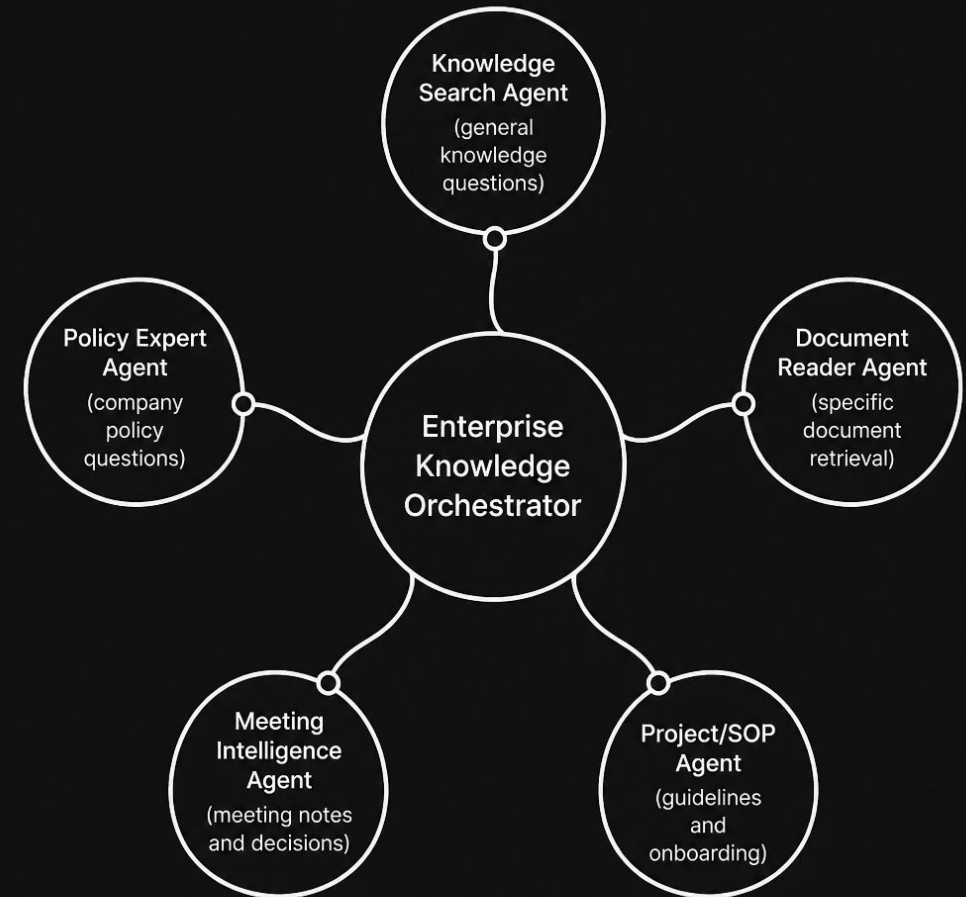


The **Enterprise Knowledge Orchestrator** serves as the central brain — interpreting user intent, selecting the right specialist agent, managing conversation memory, and ensuring all responses are grounded in verified internal knowledge.

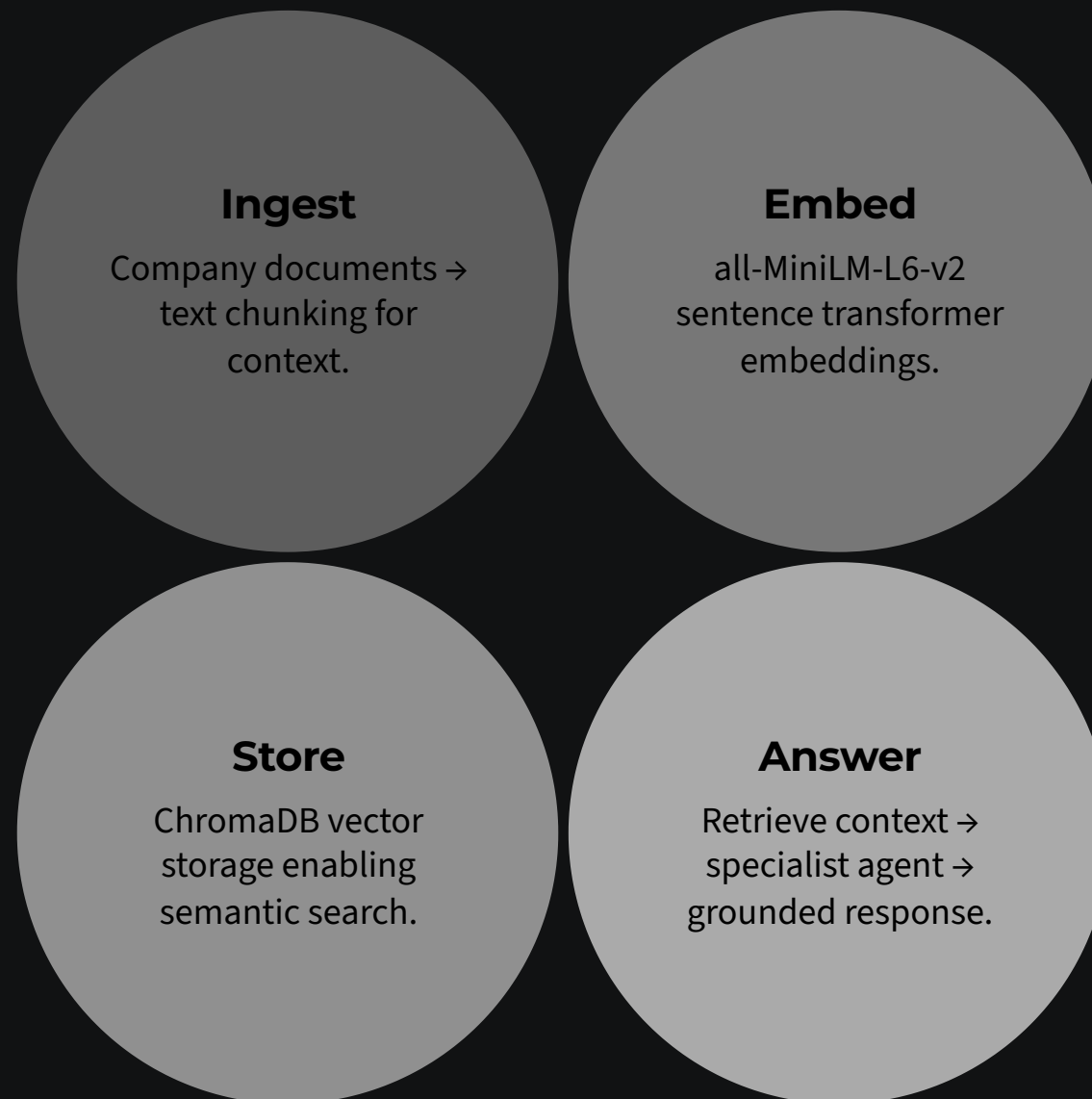
Multi-Agent Architecture

The orchestrator routes each question to the most appropriate specialist agent based on intent classification.

- **Knowledge Search Agent** — general company knowledge questions
- **Document Reader Agent** — specific document retrieval
- **Policy Expert Agent** — company policy queries
- **Meeting Intelligence Agent** — meeting notes, decisions, and action items
- **Project/SOP Agent** — project guidelines and onboarding procedures



RAG & Knowledge Retrieval Pipeline



Retrieval-Augmented Generation ensures every response is grounded in verified internal knowledge — retrieving relevant context from ChromaDB *before* generating an answer, which significantly reduces unsupported or hallucinated outputs.

Embedding Model

all-MiniLM-L6-v2

Vector Database

ChromaDB

Knowledge Categories

Policies · Meetings · Projects · SOPs · Documents

Memory, Tools & Error Handling



Conversation Memory

- Saves important interactions
- Stores question, answer, source, timestamp
- Retrieves prior context when relevant



System Tools

- Document Search & Ingestion
- Policy Search & Meeting Search
- Document Reader
- Memory Search & Save



Error Handling

- Empty input validation
- Keyboard interruption handling
- Runtime exception handling
- Missing response detection

End-to-End Workflow

Orchestrator Routing
Orchestrator detects intent and selects agent.

Semantic Retrieval
Agent searches ChromaDB and fetches context.

User Question
User asks a natural language question.

Grounded Response
Agent generates a verified answer returned to user.

Every interaction flows through the orchestrator, ensuring consistent routing, context awareness via conversation memory, and responses grounded in verified company knowledge — from question to answer.

Live Demo & Results

1

"What is the work from home policy?"

→ Policy Expert Agent

2

"What are the employee onboarding procedures?"

→ Project/SOP Agent

3

"What decisions were made in the sales meeting?"

→ Meeting Intelligence Agent

✓ End-to-End Tested

✓ Specialist Routing

✓ Knowledge Retrieval

✓ Error Handling



Conclusion & Future Scope

Conclusion

A practical enterprise AI knowledge assistant combining multi-agent architecture, RAG, vector search, tools, and memory — delivering a unified, intelligent interface for internal knowledge retrieval at NovaTech Solutions.

Future Scope

- Web-based user interface
- Authentication & role-based access control
- Larger enterprise knowledge bases
- Advanced observability & analytics
- Cloud deployment
- Additional enterprise integrations

Thank You

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